

YouTube Data Pipeline

1. Selected Data Source

YouTube

Source: YouTube Data API

Objective:

The objective is to collect video and engagement data from YouTube and use it for processing, storage, and visualization.

Data Collected

- Video ID
- Video Title
- Channel Name
- Category
- Published Date
- Views
- Likes
- Comments

2. Pipeline Overview

YouTube

↓

Ingestion

↓

Processing

↓

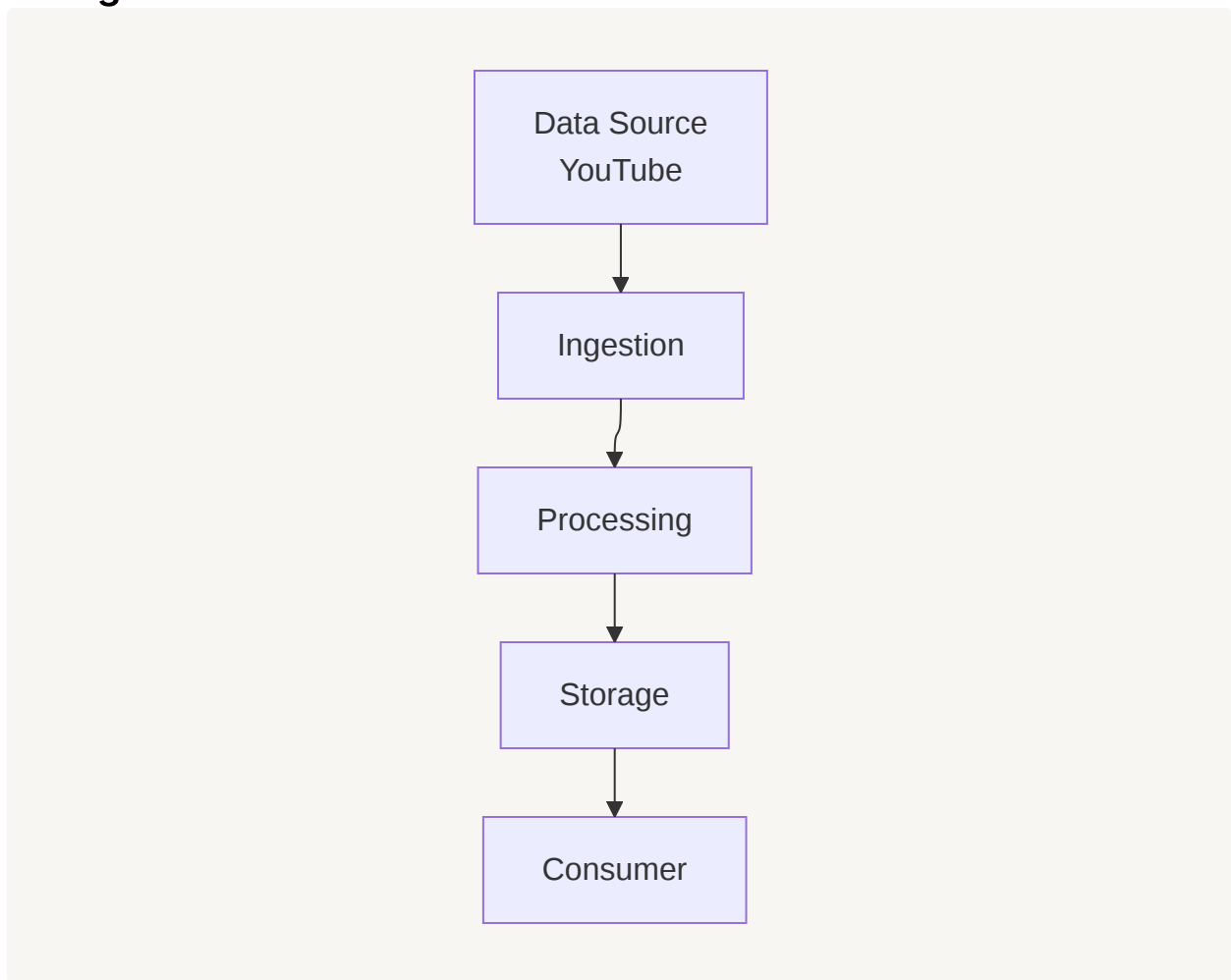
Storage



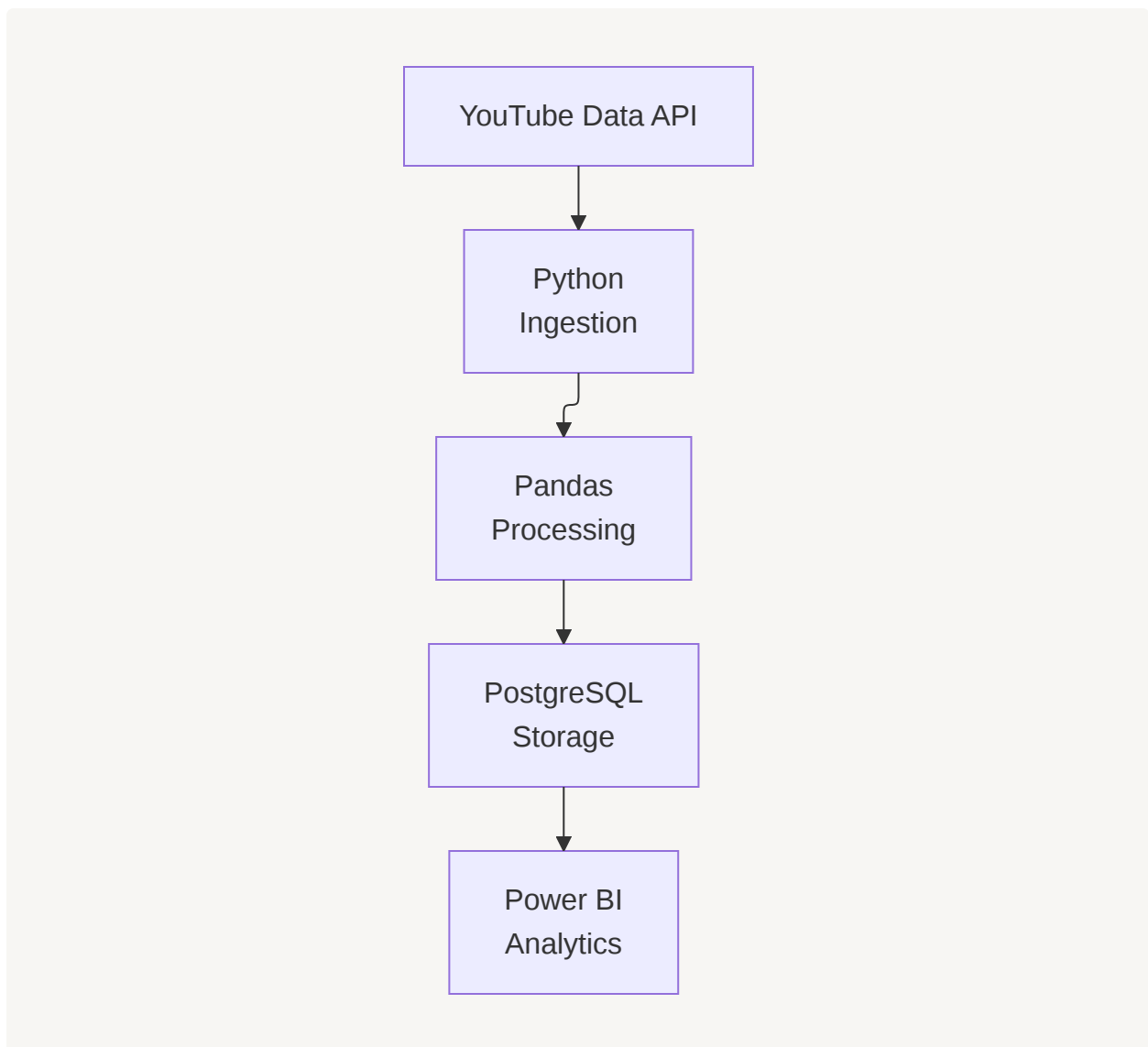
Dashboard / Analytics

The pipeline collects data from YouTube, process the collected data, store the cleaned data and finally makes it available for analytics and visualization.

3. Logical Architecture



4. Technology Mapping



5. Component Explanation

Component	Purpose
YouTube Data API	To obtain data from YouTube
Python	To collect data from API
Pandas	To clean and transform data
PostgreSQL	To store processed data
Power BI	To visualize data through dashboards and reports

6. Data Flow Explanation

YouTube → Python

Python YouTube API help to collect required data.

Python → Pandas

Collected raw data are used to clean and transform the data.

Pandas → PostgreSQL

Cleaned data is being stored into structured database.

PostgreSQL → Power BI

Power BI uses stored data to create dashboard and reports.

7. Question Log

- | Why use PostgreSQL instead of a CSV file?
- | Why do we need a processing stage?
- | Why can't the dashboard directly read data from YouTube API?

8. Architecture Decision

Why did I choose this architecture?

- | I selected a simple batch-based architecture because the objective is to collect and analyze YouTube data rather than provide real-time results. The architecture contains only the components required for data collection,

processing, storage and visualization. Technologies such as Kafka, Spark and Airflow were not included because they are not necessary for this simple use case.

9. Short Architecture Explanation

YouTube is the data source. Python collects the data through the YouTube Data API. Pandas processes and cleans the data. PostgreSQL stores the processed data, and Power BI uses the stored data for analytics and visualization.

10. Conclusion

This pipeline demonstrates how data moves from a real-world source to a final consumer. The architecture follows the basic flow of Source → Ingestion → Processing → Storage → Consumer. The design is intentionally simple so that every component has a clear purpose.

11. References

- Google Cloud — Data Pipeline Architecture
- YouTube Data API Documentation
- PostgreSQL Documentation
- Microsoft Power BI Documentation

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